

29 Beyond Perturbation Theory

29.2 Supersymmetry Current Sum Rules

We now turn to sum rules that yield exact quantitative relations between the vacuum energy and parameters describing the strength of supersymmetry breaking.

Let us again start by assuming that the world is placed in a box of volume V , with periodic boundary conditions to preserve translation invariance. The vacuum expectation value of the anticommutation relations (25.2.36) may then be expressed as a sum over discrete states $|X, \text{Box}\rangle$:

$$\begin{aligned} & \sum_X \langle \text{VAC} | Q_\alpha | X, \text{Box} \rangle \langle \text{VAC} | Q_\beta | X, \text{Box} \rangle^* \\ & + \sum_X \langle \text{VAC} | Q_\beta^* | X, \text{Box} \rangle \langle \text{VAC} | Q_\alpha^* | X, \text{Box} \rangle^* \\ & = -2i(\gamma_\mu \beta)_{\alpha\beta} \langle \text{VAC} | P^\mu | \text{VAC} \rangle. \end{aligned} \quad (29.2.1)$$

Here the label ‘Box’ indicates that the states are normalized to have Kronecker deltas rather than delta functions as their scalar products. Setting $\beta = \alpha$, summing over α , and using Eq. (25.2.37) gives

$$\sum_{X,\alpha} |\langle \text{VAC} | Q_\alpha | X, \text{Box} \rangle|^2 = 4 \langle \text{VAC} | P^0 | \text{VAC} \rangle. \quad (29.2.2)$$

Because Q_α commutes with the four-momentum, it is only states with zero three-momentum and the same energy as the vacuum that can contribute to this sum. To find the volume dependence of the matrix elements that do contribute in Eq. (29.2.2), we note that a box-normalized state $|X, \text{Box}\rangle$ containing N_X particles is related according to Eq. (3.4.3) to the corresponding continuum-normalized state $|X\rangle$ by

$$|X, \text{Box}\rangle = ((2\pi)^3/V)^{N_X/2} |X\rangle. \quad (29.2.3)$$

For states with $\mathbf{p}_X = 0$, the spatial integral of the supersymmetry current time-component S_α^0 gives another factor of V , so we conclude that for box-normalized states with $\mathbf{p}_X = 0$

$$\langle \text{VAC} | Q_\alpha | X, \text{Box} \rangle = (2\pi)^{3N_X/2} V^{1-N_X/2} \langle \text{VAC} | S_\alpha^0(0) | X \rangle. \quad (29.2.4)$$

Since invariance under 2π -rotations does not allow X to be a zero-particle state, the dominant terms in Eq. (29.2.2) when $V \rightarrow \infty$ will be those from one-particle states. In this limit, Eq. (29.2.2) becomes

$$(2\pi)^3 \sum_{X,\alpha}^{(0)} |\langle \text{VAC} | S_\alpha^0(0) | X \rangle|^2 = 4\rho_{\text{VAC}}, \quad (29.2.5)$$

where ρ_{VAC} is the vacuum energy density

$$\rho_{\text{VAC}} \equiv \langle \text{VAC} | P^0 | \text{VAC} \rangle / V, \quad (29.2.6)$$

and the superscript (0) indicates that the sum in Eq. (29.2.5) runs only over one-particle states with zero four-momentum. These of course are the two helicity states of the goldstino.

We see again from Eq. (29.2.5) that if the vacuum energy density does not vanish then the vacuum is not invariant under supersymmetry transformations, but is rather transformed into one-goldstino states. Conversely, Eq. (29.2.2) shows that if the vacuum is not invariant under supersymmetry then according to Eq. (29.2.2) its energy in a finite box cannot vanish, although it is conceivable that supersymmetry transformations might take the vacuum only into multiparticle states, in which case the vacuum energy density would vanish in the limit of large volume.

To evaluate the one-goldstino contribution in Eq. (29.2.5), we use Lorentz invariance to write the matrix element of the supersymmetry current between the vacuum and a one-goldstino state $|\mathbf{p}, \lambda\rangle$, with momentum $\mathbf{p}$ and helicity λ , in the form¹

$$\begin{aligned} \langle \text{VAC} | S^\mu(0) | \mathbf{p}, \lambda \rangle = (2\pi)^{-3/2} & \left[\left(\frac{1 + \gamma_5}{2} \right) (\gamma^\mu F + i p^\mu F') \right. \\ & \left. + \left(\frac{1 - \gamma_5}{2} \right) (\gamma^\mu F^* + i p^\mu F'^*) \right] u(\mathbf{p}, \lambda), \end{aligned} \quad (29.2.7)$$

where $u(\mathbf{p}, \lambda)$ are the coefficient functions for a massless Dirac field introduced in Section 5.5, and F and F' are unknown constants. The matrix element (29.2.7) satisfies a conservation condition $p_\mu \langle \text{VAC} | S^\mu(0) | \mathbf{p}, \lambda \rangle = 0$ because $u(\mathbf{p}, \lambda)$ satisfies the momentum-space Dirac equation (5.5.42) for zero mass, and p^μ is on the light-cone. The sum over helicities gives

$$\sum_\lambda u(\mathbf{p}, \lambda) \bar{u}(\mathbf{p}, \lambda) = -\frac{i \not{p}}{2p^0}.$$

(The Dirac spinor $u(\mathbf{p}, \lambda)$ for a massless particle of momentum $\mathbf{p}$ and spin λ is not well defined for $\mathbf{p} \rightarrow 0$, but there is no problem if we regard the sum over X in Eq. (29.2.5) as

¹Lorentz invariance alone would yield this formula with independent coefficients F_L, F'_L and F_R, F'_R for the matrices proportional to $(1 + \gamma_5)/2$ and $(1 - \gamma_5)/2$, respectively. It is CPT invariance that imposes the relations $F_R = F_L^*$ and $F'_R = F'_L^*$. To see this, we must use the CPT-transformation properties of the supersymmetry current

$$\text{CPT} S^\mu(x) (\text{CPT})^{-1} = -\gamma_5 S^\mu(-x)^* = -\beta \epsilon S^\mu(-x)$$

(see Section 5.8) and of the one-particle states

$$\text{CPT} |\mathbf{p}, \lambda\rangle = \chi_\lambda |\mathbf{p}, -\lambda\rangle$$

with χ_λ a phase factor that depends on how we define the relative phases of the helicity states. We also need the reality properties of the coefficient functions $u(\mathbf{p}, \lambda)$. These are related to those of the one-particle states by the definition

$$\langle \text{VAC} | \psi_{\text{REN}}(x) | \mathbf{p}, \lambda \rangle = (2\pi)^{-3/2} \exp(ip \cdot x) u(\mathbf{p}, \lambda),$$

where $\psi_{\text{REN}}(x)$ is a renormalized Majorana field, with CPT-transformation property

$$\text{CPT} \psi_{\text{REN}}(x) (\text{CPT})^{-1} = \gamma_5 \psi_{\text{REN}}(-x)^* = \beta \epsilon \psi_{\text{REN}}(-x),$$

which yields $u(\mathbf{p}, \lambda) = \chi_\lambda^* \beta \epsilon u^*(\mathbf{p}, -\lambda)$. Putting this together with the CPT-transformation properties of the supersymmetry current and one-particle states yields the relations $F_R = F_L^*$ and $F'_R = F'_L^*$.

a sum over helicities for a small momentum $\mathbf{p}$ of fixed direction.) After a straightforward calculation, we then have

$$4\rho_{\text{VAC}} = \text{Tr} \left\{ \left(F \frac{1+\gamma_5}{2} + F^* \frac{1-\gamma_5}{2} \right) \gamma^0 \frac{-i\not{p}}{2p^0} (\gamma^0)^\dagger \right. \\ \left. \times \left(F^* \frac{1+\gamma_5}{2} + F \frac{1-\gamma_5}{2} \right) \right\},$$

and therefore

$$\rho_{\text{VAC}} = |F|^2/2. \quad (29.2.8)$$

An alternative proof of this formula will be given at the end of this section.

The parameter F plays much the same role in the interactions of soft goldstinos as does the parameter F_π (introduced in Section 19.4) in the interactions of soft pions. The matrix element of the supersymmetry current between any two states X and Y may be split into terms that have a one-goldstino pole at $p^\mu = 0$ in the momentum transfer $p \equiv p_X - p_Y$ and terms that do not have this pole:

$$\langle X|S^\mu(0)|Y\rangle = \left\{ \begin{aligned} &\left(\frac{1+\gamma_5}{2} \right) [\gamma^\mu F + ip^\mu F'] \\ &+ \left(\frac{1-\gamma_5}{2} \right) [\gamma^\mu F^* + ip^\mu F'^*] \end{aligned} \right\} \left(\frac{-i\not{p}}{p^2} \right) M(X \rightarrow Y + g) \quad (29.2.9) \\ + \langle X|S^\mu(0)|Y\rangle_{\text{no pole}},$$

where $\bar{u}M(X \rightarrow Y + g)$ is the amplitude for emitting a goldstino with four-momentum p and Dirac wave function u , and the subscript ‘no pole’ denotes the terms in the matrix element that do not have the one-goldstino pole in the four-momentum p . The conservation of the current S^μ tells us that this vanishes if contracted with p_μ , so in the limit $p^\mu \rightarrow 0$ the goldstino emission amplitude is²

$$M(X \rightarrow Y + g) \xrightarrow{p^\mu \rightarrow 0} -i \left\{ \left(\frac{1+\gamma_5}{2F} \right) + \left(\frac{1-\gamma_5}{2F^*} \right) \right\} p_\mu \langle X|S^\mu(0)|Y\rangle_{\text{no pole}}. \quad (29.2.10)$$

There is another sum rule that provides an alternative proof of the existence of goldstinos when supersymmetry is spontaneously broken, and that also relates the parameter F and the vacuum energy density to the D -terms and $\mathcal{F}$ -terms that characterize the strength of supersymmetry breaking. (This use of a sum rule is analogous to the second proof in Section 19.2 of the existence of Goldstone bosons when ordinary symmetries are spontaneously broken.) To derive this sum rule, we will now give up the device of a finite volume,

²The amplitude $\langle X|S^\mu(0)|Y\rangle_{\text{no pole}}$ may have poles that go as $1/p \cdot k$ as $p^\mu \rightarrow 0$; they would not arise from the goldstino propagator, which is explicitly excluded in this matrix element, but rather from other particle propagators produced by the insertion of the supersymmetry current in external lines of momentum k of the process $X \rightarrow Y$. In the limit $p^\mu \rightarrow 0$, the contribution of these poles in Eq. (29.2.10) would dominate the amplitudes for emission or absorption of soft goldstinos. But for such poles to arise, the goldstino would have to be emitted in a transition between a pair of degenerate particles of opposite statistics, which are not likely to appear in theories in which supersymmetry is spontaneously broken. The interactions of soft goldstinos differ in this respect from the interactions of soft pions, photons, or gravitons.

and instead avoid the question of the convergence of the integral (29.1.3) by working with a local consequence of the supersymmetry of the action

$$[(\bar{S}^0(\mathbf{x}, t)\alpha), \chi(\mathbf{y}, t)] = [(\bar{\alpha}S^0(\mathbf{x}, t)), \chi(\mathbf{y}, t)] = i\delta^3(\mathbf{x} - \mathbf{y})\delta\chi(\mathbf{x}, t) + \cdots, \quad (29.2.11)$$

where $\chi(x)$ is an arbitrary fermionic or bosonic field, $\delta\chi(x)$ is the change in $\chi(x)$ induced by the supersymmetry transformation with infinitesimal parameter α that leaves the action invariant, and the dots denote terms involving derivatives of $\delta^3(\mathbf{x} - \mathbf{y})$.

Let us consider the vacuum expectation value of the anticommutator of an arbitrary left-handed spinor field $\psi_L(x)$ with the covariant conjugate $\bar{S}^\mu(y)$ of the supersymmetry current. By summing over a complete set $|X\rangle$ of intermediate states (including integration over particle momenta), this vacuum expectation value may be put in the form

$$\begin{aligned} & \langle \text{VAC} | \{\psi_{L\alpha}(x), \bar{S}_\beta^\mu(y)\} | \text{VAC} \rangle \\ &= \int d^4p \exp(ip \cdot (x - y)) \left[G_{\alpha\beta}^\mu(p) + \tilde{G}_{\alpha\beta}^\mu(-p) \right], \end{aligned} \quad (29.2.12)$$

where

$$G_{\alpha\beta}^\mu(p) \equiv \sum_X \delta^4(p - p_X) \langle \text{VAC} | \psi_{L\alpha}(0) | X \rangle \langle X | \bar{S}_\beta^\mu(0) | \text{VAC} \rangle, \quad (29.2.13)$$

$$\tilde{G}_{\alpha\beta}^\mu(p) \equiv \sum_X \delta^4(p - p_X) \langle \text{VAC} | \bar{S}_\beta^\mu(0) | X \rangle \langle X | \psi_{L\alpha}(0) | \text{VAC} \rangle. \quad (29.2.14)$$

Lorentz invariance requires that the matrices $G^\mu(p)$ and $\tilde{G}^\mu(p)$ must take the forms:

$$\begin{aligned} G^\mu(p) = \theta(p^0) \left(\frac{1 + \gamma_5}{2} \right) & \left[\gamma^\mu G^{(1)}(-p^2) + \not{p} p^\mu G^{(2)}(-p^2) \right. \\ & \left. + p^\mu G^{(3)}(-p^2) + \not{p} \gamma^\mu G^{(4)}(-p^2) \right] \end{aligned} \quad (29.2.15)$$

and

$$\begin{aligned} \tilde{G}^\mu(p) = \theta(p^0) \left(\frac{1 + \gamma_5}{2} \right) & \left[\gamma^\mu \tilde{G}^{(1)}(-p^2) + \not{p} p^\mu \tilde{G}^{(2)}(-p^2) \right. \\ & \left. + p^\mu \tilde{G}^{(3)}(-p^2) + \not{p} \gamma^\mu \tilde{G}^{(4)}(-p^2) \right]. \end{aligned} \quad (29.2.16)$$

The right-hand sides of Eqs. (29.2.15) and (29.2.16) are unaffected if we replace $-p^2$ with m^2 , multiply by a factor $\delta(p^2 + m^2)$, and integrate over m^2 . In this way, Eq. (29.2.12) becomes

$$\begin{aligned} & (2\pi)^{-3} \langle \text{VAC} | \{\psi_L(x), \bar{S}^\mu(y)\} | \text{VAC} \rangle \\ &= \left(\frac{1 + \gamma_5}{2} \right) \int_0^\infty dm^2 \left[\begin{aligned} & \gamma^\mu G^{(1)}(m^2) - \not{\partial} \partial^\mu G^{(2)}(m^2) \\ & - i \partial^\mu G^{(3)}(m^2) - i \not{\partial} \gamma^\mu G^{(4)}(m^2) \end{aligned} \right] \Delta_+(x - y, m) \\ &+ \left(\frac{1 + \gamma_5}{2} \right) \int_0^\infty dm^2 \left[\begin{aligned} & \gamma^\mu \tilde{G}^{(1)}(m^2) - \not{\partial} \partial^\mu \tilde{G}^{(2)}(m^2) \\ & - i \partial^\mu \tilde{G}^{(3)}(m^2) - i \not{\partial} \gamma^\mu \tilde{G}^{(4)}(m^2) \end{aligned} \right] \Delta_+(y - x, m), \end{aligned} \quad (29.2.17)$$

where $\Delta_+(x, m)$ is the standard function

$$\Delta_+(x, m) \equiv (2\pi)^{-3} \int d^4p \theta(p^0) \delta(p^2 + m^2) \exp(ip \cdot x). \quad (29.2.18)$$

Causality requires that the anticommutator on the left-hand side of Eq. (29.2.17) should vanish for spacelike separations $x-y$. For such separations, $\Delta_+(x-y, m)$ is an even function of $x-y$, so Eq. (29.2.17) vanishes for all spacelike separations if and only if

$$\begin{aligned} G^{(1)}(m^2) &= -\tilde{G}^{(1)}(m^2), & G^{(2)}(m^2) &= -\tilde{G}^{(2)}(m^2), \\ G^{(3)}(m^2) &= +\tilde{G}^{(3)}(m^2), & G^{(4)}(m^2) &= +\tilde{G}^{(4)}(m^2), \end{aligned} \quad (29.2.19)$$

so that for general $x-y$, Eq. (29.2.17) now reads

$$\begin{aligned} & (2\pi)^{-3} \langle \text{VAC} | \{ \psi_L(x), \bar{S}^\mu(y) \} | \text{VAC} \rangle \\ &= \int_0^\infty dm^2 \left[\begin{aligned} & G^{(1)}(m^2) \gamma^\mu - G^{(2)}(m^2) \not{\partial} \partial^\mu \\ & - i G^{(3)}(m^2) \partial^\mu - i G^{(4)}(m^2) \not{\partial} \gamma^\mu \end{aligned} \right] \Delta(x-y, m) \left(\frac{1+\gamma_5}{2} \right), \end{aligned} \quad (29.2.20)$$

where, as usual,

$$\Delta(x-y, m) \equiv \Delta_+(x-y, m) - \Delta_+(y-x, m). \quad (29.2.21)$$

We next impose the supersymmetry current conservation condition (29.1.2), which (because $\square \Delta = m^2 \Delta$) yields

$$G^{(1)}(m^2) = m^2 G^{(2)}(m^2), \quad m^2 G^{(3)}(m^2) = -m^2 G^{(4)}(m^2). \quad (29.2.22)$$

Finally, we must relate these spectral functions to the breaking of supersymmetry. Recall that for $x^0 = y^0$,

$$\Delta(\mathbf{x} - \mathbf{y}, 0, m) = 0, \quad \dot{\Delta}(\mathbf{x} - \mathbf{y}, 0, m) = -i\delta^3(\mathbf{x} - \mathbf{y}).$$

Setting $x^0 = y^0 = t$ and $\mu = 0$ in Eq. (29.2.20) thus yields

$$\begin{aligned} & \langle \text{VAC} | \{ \psi_L(\mathbf{x}, t), \bar{S}^0(\mathbf{y}, t) \} | \text{VAC} \rangle \\ &= (2\pi)^3 \delta^3(\mathbf{x} - \mathbf{y}) \left(\frac{1+\gamma_5}{2} \right) \int_0^\infty dm^2 \left[G^{(3)}(m^2) + G^{(4)}(m^2) \right]. \end{aligned} \quad (29.2.23)$$

Contracting on the right with an infinitesimal Majorana fermionic supersymmetry transformation parameter α and using Eq. (29.2.11) then gives

$$i \langle \delta \psi_L \rangle_{\text{VAC}} = (2\pi)^3 \left(\frac{1+\gamma_5}{2} \right) \alpha \int_0^\infty dm^2 \left[G^{(3)}(m^2) + G^{(4)}(m^2) \right]. \quad (29.2.24)$$

But Eq. (29.2.22) shows that the integrand of the integral over m^2 in Eq. (29.2.24) vanishes, except perhaps at $m^2 = 0$, so we can conclude that

$$G^{(3)}(m^2) + G^{(4)}(m^2) = \delta(m^2) \mathcal{G}, \quad (29.2.25)$$

with a constant coefficient $\mathcal{G}$ given by

$$i \langle \delta \psi_L \rangle_{\text{VAC}} = (2\pi)^3 \mathcal{G} \alpha_L. \quad (29.2.26)$$

As we saw in Sections 26.4 and 27.4, a breakdown of supersymmetry is signaled by the appearance of vacuum expectation values of the changes $\delta\psi$ under supersymmetry

transformations of one or more spinor fields ψ . Eqs. (29.2.25) and (29.2.26) show that for any such spinor field, the spectral function $G^{(3)}(m^2) + G^{(4)}(m^2)$ has a delta function singularity at $m^2 = 0$, which could only arise from the appearance of a massless one-particle state $|g\rangle$ in the sums over states in Eqs. (29.2.13) and/or (29.2.14). For the matrix elements $\langle \text{VAC} | \psi | g \rangle$ or $\langle g | \psi | \text{VAC} \rangle$ not to vanish, this massless particle must have spin 1/2. This is the goldstino.

Now let's calculate the contribution of a one-goldstino state $|\mathbf{p}, \lambda\rangle$ with momentum $\mathbf{p}$ and helicity λ to the spectral functions $G^{(i)}(m^2)$. Lorentz invariance tells us that the matrix element of a general fermion field (not just the renormalized goldstino field) in Eq. (29.2.13) takes the form³

$$\langle \text{VAC} | \psi(0) | \mathbf{p}, \lambda \rangle = \frac{1}{(2\pi)^{3/2}} \left[N \left(\frac{1 + \gamma_5}{2} \right) + N^* \left(\frac{1 - \gamma_5}{2} \right) \right] u(\mathbf{p}, \lambda), \quad (29.2.27)$$

where N is a constant, characterizing the particular fermion field. Also, the delta function in Eq. (29.2.13) is

$$\delta^4(p - p_g) = 2p^0 \delta(p^2) \theta(p^0) \delta^3(\mathbf{p} - \mathbf{p}_g).$$

Together with Eq. (29.2.7), this yields

$$\left[G_{\alpha\beta}^\mu(p) \right]_1 = \frac{N}{(2\pi)^3} \left(\frac{1 + \gamma_5}{2} \right) \theta(p^0) \delta(p^2) \not{p} [\gamma^\mu F + i p^\mu F'],$$

The subscript 1 indicates the one-goldstino contribution. Comparison with Eq. (29.2.15) shows that

$$\begin{aligned} \left[G^{(2)}(m^2) \right]_1 &= i(2\pi)^{-3} N F' \delta(m^2), & \left[G^{(4)}(m^2) \right]_1 &= (2\pi)^{-3} N F \delta(m^2), \\ \left[G^{(1)}(m^2) \right]_1 &= \left[G^{(3)}(m^2) \right]_1 = 0. \end{aligned} \quad (29.2.28)$$

Thus Eqs. (29.2.25) and (29.2.26) yield

$$i \langle \delta \psi_L \rangle_{\text{VAC}} = N F \alpha_L. \quad (29.2.29)$$

Evidently if the change in any fermion field under a supersymmetry transformation has a non-zero vacuum expectation value then the N -factor for that field cannot vanish, so there must be a one-goldstino state that contributes to these spectral functions.

To be a little more specific, recall that the fermionic components ψ_{Ln} of a left-chiral scalar superfield Φ_{Ln} obey the supersymmetry transformation rule (26.3.15):

$$\delta \psi_{Ln} = \sqrt{2} \partial_\mu \phi_n \gamma^\mu \alpha_R + \sqrt{2} \mathcal{F}_n \alpha_L, \quad (29.2.30)$$

while Eq. (27.3.5) gives the supersymmetry transformation of the gaugino fields as

$$\delta \lambda_A = \left(\frac{1}{4} f_{A\mu\nu} [\gamma^\nu, \gamma^\mu] + i \gamma_5 D_A \right) \alpha. \quad (29.2.31)$$

³Again, Lorentz invariance would allow independent coefficients N_L and N_R for $(1 + \gamma_5)$ and $(1 - \gamma_5)$, respectively. Using the CPT transformation of the one-particle state and the reality property of the coefficient functions discussed in the first footnote of this section, and the CPT transformation of a general fermion field, $\text{CPT } \psi(x) (\text{CPT})^{-1} = \beta \epsilon \psi(-x)$, we find that $N_L = N_R^*$.

Hence the N -factors in the matrix elements of the ψ_n and λ_A between the vacuum and a one-goldstino state are given by

$$N_n = iF^{-1}\sqrt{2}\langle\mathcal{F}_n\rangle_{\text{VAC}}, \quad N_A = -F^{-1}\langle D_A\rangle_{\text{VAC}}. \quad (29.2.32)$$

Let us see how the results (29.2.32) arise in the tree approximation. For a renormalizable theory of gauge and chiral superfields, Eq. (27.4.30) gives the left-handed part of the fermion mass matrix as

$$M_{nm} = \left(\frac{\partial^2 f(\phi)}{\partial\phi_n \partial\phi_m} \right)_{\phi=\phi_0}, \quad (29.2.33)$$

$$M_{nA} = M_{An} = i\sqrt{2}(t_A\phi_0)_n^*, \quad M_{AB} = 0.$$

The vacuum fields ϕ_{n0} are at a minimum of the potential given by Eq. (27.4.9), so

$$0 = \left(\frac{\partial V(\phi)}{\partial\phi_n} \right)_0 = -\sum_m M_{nm}\mathcal{F}_{m0} + \sum_A (\phi_0^\dagger t_A)_n D_{A0}, \quad (29.2.34)$$

where the $\mathcal{F}$ s and D s are given by Eqs. (27.4.6) and (27.4.7)

$$\mathcal{F}_n = -(\partial f(\phi)/\partial\phi_n)^*, \quad D_A = \xi_A + \sum_{nm} \phi_n^*(t_A)_{nm}\phi_m,$$

and the subscript 0 indicates that we are to set $\phi_n = \phi_{n0}$. Furthermore, the gauge invariance of the superpotential requires that

$$\sum_n \mathcal{F}_n (t_B\phi)_n^* = 0, \quad (29.2.35)$$

for all values of ϕ . Therefore the left-handed quark mass matrix M has an eigenvector v with $Mv = 0$, where

$$v_n = \sqrt{2}\mathcal{F}_{n0}, \quad v_A = iD_{A0}. \quad (29.2.36)$$

Thus in the tree approximation, if we expand the left-handed fermion fields in renormalized fields for particles of definite mass, then the coefficients of the goldstino field in ψ_{Ln} and in λ_{AL} are proportional to $\sqrt{2}\mathcal{F}_{n0}$ and iD_{A0} , respectively, in agreement with Eq. (29.2.32).

If we normalize the spinor fields ψ_{Ln} and λ_A so that the matrix connecting these fields to the renormalized fields of particles of definite mass is unitary, then

$$\sum_n |N_n|^2 + \sum_A |N_A|^2 = 1, \quad (29.2.37)$$

so Eq. (29.2.32) gives the non-perturbative result

$$|F|^2 = 2 \sum_n |\langle\mathcal{F}_n\rangle_{\text{VAC}}|^2 + \sum_A |\langle D_A\rangle_{\text{VAC}}|^2. \quad (29.2.38)$$

The result (29.2.38) for $|F|^2$ allows us to express the vacuum expectation density (29.2.8) in terms of the vacuum expectation values of auxiliary fields

$$\rho_{\text{VAC}} = \sum_n |\langle\mathcal{F}_n\rangle_{\text{VAC}}|^2 + \frac{1}{2} \sum_A |\langle D_A\rangle_{\text{VAC}}|^2. \quad (29.2.39)$$

This is the non-perturbative generalization of the zeroth-order result (27.4.9). It confirms a result used in Section 27.6, that the conditions $\langle \mathcal{F}_n \rangle = \langle D_A \rangle = 0$ are sufficient as well as necessary for supersymmetry to be unbroken.

* * *

It is instructive to see how Eq. (29.2.8) can be derived without the device of a finite volume. For this purpose, consider the vacuum expectation value of the anticommutator of two supersymmetry currents. By using Lorentz invariance and the vanishing of anticommutators at spacelike separations in the same way as in the previous section, we find

$$\begin{aligned} \langle \text{VAC} | \{S^\mu(x), \bar{S}^\nu(y)\} | \text{VAC} \rangle \\ = -i \int dm^2 [H^{(1)}(m^2) \gamma^\mu \partial^\nu + H^{(2)}(m^2) \gamma^\nu \partial^\mu + H^{(3)}(m^2) \not{\partial} \partial^\mu \partial^\nu \\ + H^{(4)}(m^2) \not{\partial} \eta^{\mu\nu} + H^{(5)}(m^2) \epsilon^{\mu\nu\lambda\rho} \partial_\lambda \gamma_\rho] \Delta(x-y, m^2) + \dots \end{aligned} \quad (29.2.40)$$

Here

$$\begin{aligned} \int dX \delta^4(p - p_X) \langle \text{VAC} | S_\alpha^\mu(0) | X \rangle \langle X | \bar{S}_\beta^\nu(0) | \text{VAC} \rangle \\ = H^{(1)}(-p^2) \gamma^\mu p^\nu + H^{(2)}(-p^2) \gamma^\nu p^\mu - H^{(3)}(-p^2) \not{p} p^\mu p^\nu + H^{(4)}(-p^2) \not{p} \eta^{\mu\nu} \\ + H^{(5)}(-p^2) \epsilon^{\mu\nu\lambda\rho} p_\lambda \gamma_\rho + \dots, \end{aligned} \quad (29.2.41)$$

and the dots in Eqs. (29.2.40) and (29.2.41) denote a linear combination of the other independent Dirac covariant matrices $\mathbf{1}$, γ_5 , $\gamma_5 \gamma_\sigma$, and $[\gamma_\sigma, \gamma_\tau]$, which will not concern us here. The Majorana nature of the currents together with Eq. (26.A.20) tell us that the spectral functions $H^{(i)}$ are all real, and the conservation of the supersymmetry currents dictate that

$$H^{(1)}(m^2) = H^{(2)}(m^2) = -m^2 H^{(3)}(m^2) - H^{(4)}(m^2), \quad (29.2.42)$$

and

$$m^2 H^{(1)}(m^2) = 0. \quad (29.2.43)$$

Setting $\mu = \nu = 0$ and $x^0 = y^0$ in Eq. (29.2.40), integrating over $\mathbf{x}$, and using Eq. (29.2.42) then gives

$$\begin{aligned} \langle \text{VAC} | \{Q, \bar{S}^0(0)\} | \text{VAC} \rangle &= (2\pi)^3 \beta \int dm^2 [H^{(1)}(m^2) + H^{(2)}(m^2) \\ &\quad + m^2 H^{(3)}(m^2) + H^{(4)}(m^2)] + \dots \\ &= (2\pi)^3 \beta \int dm^2 H^{(1)}(m^2) + \dots \end{aligned} \quad (29.2.44)$$

To evaluate this anticommutator, we first write it as

$$\{Q, \bar{S}^\nu(x)\} = -2i \gamma_\mu T^{\mu\nu}(x) + \dots, \quad (29.2.45)$$

where $T^{\mu\nu}(x)$ is some tensor operator and the dots again denote some linear combination of the other independent Dirac covariants. The Majorana character of Q and S^μ tells us

that $T^{\mu\nu}(x)$ is Hermitian; the conservation of the supersymmetry current tells us that it is conserved, in the sense that

$$\partial_\nu T^{\mu\nu} = 0, \quad (29.2.46)$$

and the anticommutation relation (25.2.36) tells us that

$$\int d^3x T^{\mu 0}(x) = P^\mu. \quad (29.2.47)$$

These properties allow us to identify $T^{\mu\nu}$ as the energy-momentum tensor. It is not in general equal to the symmetric energy-momentum tensor $\Theta^{\mu\nu}$ discussed in Section 7.4, but Eq. (29.2.47) shows that the energy density T^{00} can differ from Θ^{00} only by spatial derivative terms that cannot contribute in states of zero three-momentum, so the vacuum energy density is given by

$$\rho_{\text{VAC}} = \langle \text{VAC} | T^{00} | \text{VAC} \rangle. \quad (29.2.48)$$

Thus Eqs. (29.2.44) and (29.2.45) yield

$$2\rho_{\text{VAC}} = (2\pi)^3 \int dm^2 H^{(1)}(m^2). \quad (29.2.49)$$

But Eq. (29.2.43) tells us that $H^{(1)}(m^2)$ vanishes except perhaps at $m^2 = 0$, so

$$H^{(1)}(m^2) = 2(2\pi)^{-3} \delta(m^2) \rho_{\text{VAC}}. \quad (29.2.50)$$

Thus we see again that a non-vanishing vacuum energy density entails the existence of a massless fermion, the goldstino. Using Eq. (29.2.7), a straightforward calculation of the one-goldstino contribution to the spectral functions yields

$$H^{(1)}(m^2) = (2\pi)^{-3} \delta(m^2) |F|^2. \quad (29.2.51)$$

Comparing this with Eq. (29.2.50) gives our previous result (29.2.8) for the vacuum energy density.